RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Friday, 25 September 2026, 2:31 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:51 PM
Time taken	20 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    int A[n];
    // Read the array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    // Read the target difference k
    scanf("%d", &k);

    int found = 0;

    // O(n^2) nested loop approach to find the pair
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            // Check condition: i != j and A[j] - A[i] == k
            if (i != j && (A[j] - A[i] == k)) {
                found = 1;
                break;
            }
        }
        if (found) {
            break;
        }
    }

    // Output 1 if pair exists, 0 otherwise
    printf("%d\n", found);

    return 0;
```

```
40 }
41
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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